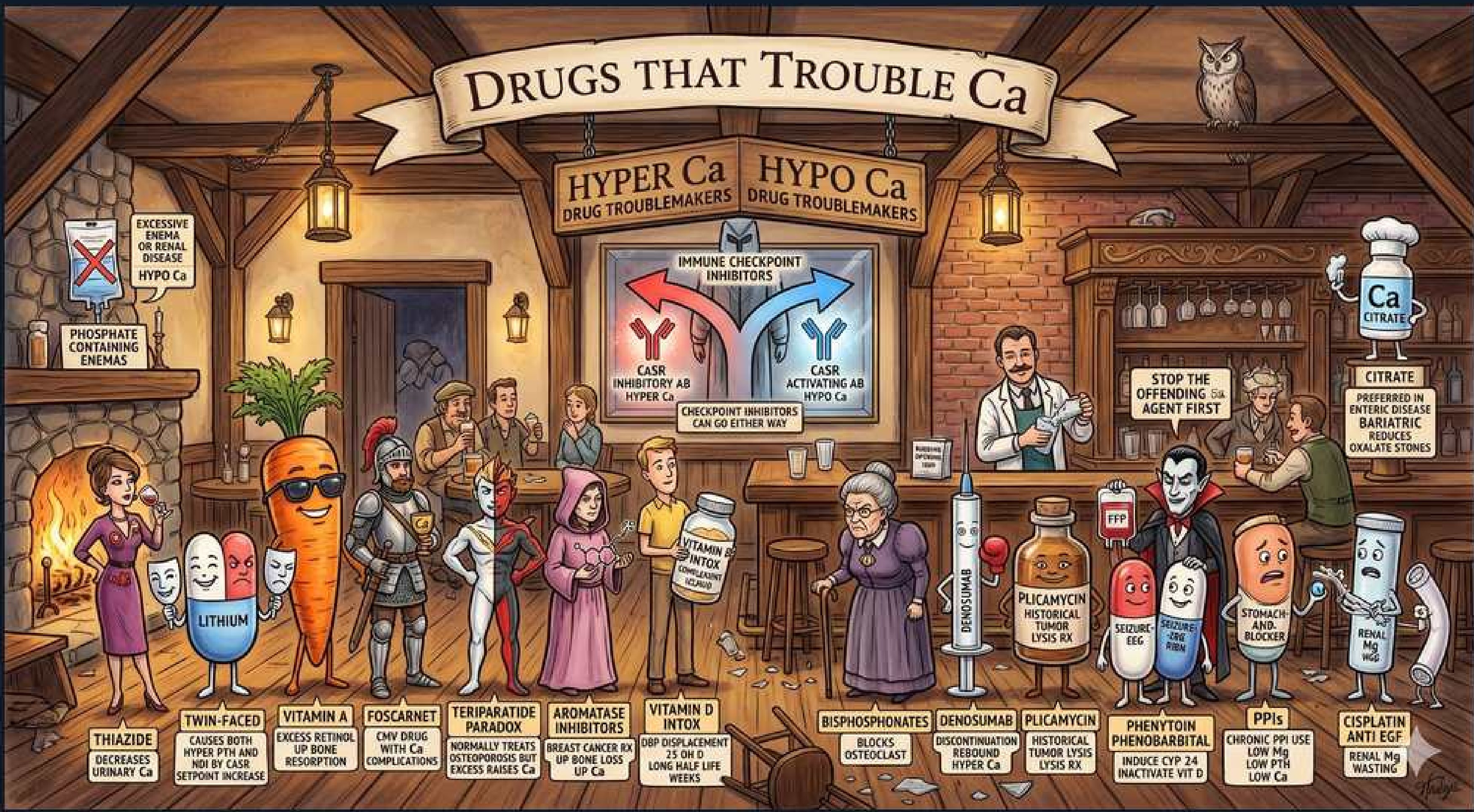
Calcium (Renal) — Drugs



1. Drugs that trouble Ca

WHAT YOU SEE

Banner: DRUGS THAT TROUBLE Ca

WHAT IT ENCODES

Mnemonic tavern: the LEFT side of the bar holds agents that RAISE calcium (hypercalcemia), the RIGHT side holds agents that LOWER it (hypocalcemia). On boards, a drug history is mandatory in any unexplained calcium abnormality — discontinuing the culprit is often the entire treatment. Always reconcile thiazides, lithium, vitamin D/A, antiresorptives, anticonvulsants, PPIs, and citrate exposure.

BOARD TRAP

WRONG: launching a full malignancy/parathyroid workup before reviewing the medication list. A thiazide or lithium can fully explain mild hypercalcemia, and bisphosphonate/denosumab or citrate can explain hypocalcemia — stopping the drug comes first.

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2. HyperCa troublemakers

WHAT YOU SEE

Left arch: HYPER Ca drug troublemakers

WHAT IT ENCODES

The hypercalcemia-causing drugs: thiazide diuretics (decrease urinary Ca), lithium (raises PTH set point), excess vitamin A (osteoclastic bone resorption), teriparatide (intermittent PTH analog), and vitamin D intoxication (increased gut absorption). Antacid/calcium + alkali (milk-alkali) also belongs here. Most cause MILD hypercalcemia that resolves on withdrawal.

BOARD TRAP

WRONG: attributing severe symptomatic hypercalcemia (>14) to a thiazide alone. Drug-induced hypercalcemia is usually mild; a markedly high calcium points to primary hyperPTH or malignancy that the drug merely unmasked.

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3. HypoCa troublemakers

WHAT YOU SEE

Right arch: HYPO Ca drug troublemakers

WHAT IT ENCODES

The hypocalcemia-causing drugs: bisphosphonates and denosumab (block osteoclasts), plicamycin, phenytoin/phenobarbital (induce CYP → degrade vitamin D), chronic PPIs (reduce Ca absorption), citrate (chelation in massive transfusion/CRRT), and foscarnet/phosphate enemas (chelate/bind Ca). Antiresorptives are the highest-yield: always check and replete vitamin D BEFORE giving them.

BOARD TRAP

WRONG: giving a bisphosphonate or denosumab to a patient with unrecognized vitamin D deficiency. This precipitates severe, prolonged hypocalcemia (especially with denosumab) — correct 25-D and calcium first.

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4. Checkpoint inhibitors both

WHAT YOU SEE

Immune checkpoint inhibitors, CaSR arrows both ways

WHAT IT ENCODES

Immune checkpoint inhibitors (anti-PD-1/PD-L1, anti-CTLA-4) can swing calcium EITHER way: immune hypophysitis/adrenalitis or hypoparathyroidism (via anti-CaSR-activating autoantibodies) → hypocalcemia, while disease flares or sarcoid-like granulomatous reactions → hypercalcemia. With rising cancer immunotherapy use, this is an increasingly tested cause of endocrinopathy.

BOARD TRAP

WRONG: assuming a checkpoint-inhibitor patient's calcium derangement is always tumor-related (PTHrP/lytic). Autoimmune hypoparathyroidism and granulomatous reactions are drug-induced and reversible/steroid-responsive — check PTH and the autoimmune panel.

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5. Thiazide ↑Ca

WHAT YOU SEE

Thiazide woman, decreases urinary Ca

WHAT IT ENCODES

Thiazides act at the distal convoluted tubule and INCREASE distal calcium reabsorption, lowering URINARY calcium and modestly raising serum calcium — which is exactly why they are used to prevent calcium kidney stones. They can also UNMASK underlying primary hyperPTH. Hold the thiazide and recheck; if hypercalcemia persists, work up for hyperPTH.

BOARD TRAP

WRONG: confusing thiazide and loop effects on calcium. Thiazides RETAIN calcium (hypercalcemia, used for stone prevention); loops (furosemide) DUMP calcium in urine. 'Loops Lose calcium' is the discriminator.

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6. Lithium ↑Ca

WHAT YOU SEE

Two-faced lithium pill, raises PTH set point

WHAT IT ENCODES

Lithium raises the calcium set point of the CaSR on parathyroid cells, so higher calcium is needed to suppress PTH → mild hypercalcemia with high/normal PTH (a phenocopy of FHH or primary hyperPTH). It can also cause nephrogenic diabetes insipidus. Chronic use may lead to true parathyroid hyperplasia/adenoma.

BOARD TRAP

WRONG: diagnosing primary hyperparathyroidism in a lithium-treated bipolar patient without considering lithium itself. The labs look identical (high Ca, high/normal PTH); the drug history is the discriminator — trial of lithium cessation if feasible.

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7. Vitamin A ↑Ca

WHAT YOU SEE

Carrot character, excess vitamin A, bone resorption

WHAT IT ENCODES

Hypervitaminosis A (high-dose supplements, isotretinoin/retinoids) stimulates osteoclastic bone resorption → hypercalcemia, plus pseudotumor cerebri, dry skin, and hepatotoxicity. Treatment is stopping the vitamin; hydration and rarely a short steroid course.

BOARD TRAP

WRONG: overlooking retinoids/supplements as a cause and chasing malignancy. A patient on isotretinoin or megadose vitamin A with bone pain and headache has a clear drug etiology — review the supplement list.

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8. Teriparatide ↑Ca

WHAT YOU SEE

Knight teriparatide, intermittent PTH

WHAT IT ENCODES

Teriparatide (PTH 1-34) and abaloparatide are ANABOLIC osteoporosis drugs: INTERMITTENT PTH stimulates osteoblasts (net bone formation), but can cause transient hypercalcemia/hypercalciuria. They carry a (rodent) osteosarcoma boxed warning and are avoided in Paget, prior skeletal radiation, or unexplained high alk phos. Limit course to ~2 years.

BOARD TRAP

WRONG: thinking PTH always resorbs bone. CONTINUOUS PTH (hyperparathyroidism) is catabolic and thins bone, but INTERMITTENT/daily teriparatide is ANABOLIC and builds bone — the dosing pattern, not the molecule, determines the effect.

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9. Vitamin D intox ↑Ca

WHAT YOU SEE

Vitamin D bottle, long half-life

WHAT IT ENCODES

Vitamin D intoxication (megadose supplements) raises gut calcium and phosphate absorption → hypercalcemia AND hyperphosphatemia, with markedly elevated 25-OH-D (the form to measure) and suppressed PTH. Because vitamin D is fat-soluble with a long half-life, hypercalcemia can persist for WEEKS. Treat with hydration, steroids, and stopping the supplement.

BOARD TRAP

WRONG: ordering 1,25-D to diagnose vitamin D toxicity. Exogenous toxicity elevates 25-OH-D (storage form); 1,25-D is often normal because its synthesis is feedback-suppressed. Measure 25-D for intoxication.

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10. Bisphosphonates ↓Ca

WHAT YOU SEE

Bisphosphonate syringe, blocks osteoclasts

WHAT IT ENCODES

Bisphosphonates (IV zoledronic acid 4 mg, pamidronate 60–90 mg) inhibit osteoclast farnesyl pyrophosphate synthase → durable suppression of bone resorption; they are the mainstay for malignant hypercalcemia (peak effect 2–4 days). Overshoot or pre-existing vitamin D deficiency causes hypocalcemia. Watch for renal toxicity, jaw osteonecrosis, and atypical femur fractures.

BOARD TRAP

WRONG: expecting a bisphosphonate to fix hypercalcemia within hours. Onset is delayed 2–4 days — that's why you bridge with IV saline and CALCITONIN for the first 48 h. And dose-reduce/avoid in significant renal impairment.

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11. Denosumab ↓Ca

WHAT YOU SEE

Denosumab vial, RANKL inhibitor

WHAT IT ENCODES

Denosumab is a monoclonal anti-RANKL antibody that blocks osteoclast formation/activity, lowering calcium; it is the preferred antiresorptive for hypercalcemia in RENAL FAILURE (not cleared by kidney, no dose adjustment) and for bisphosphonate-refractory cases. It causes more frequent and severe hypocalcemia than bisphosphonates, especially with low vitamin D or CKD.

BOARD TRAP

WRONG: giving denosumab without first repleting calcium/vitamin D in a CKD patient. This is a classic cause of life-threatening hypocalcemia — and stopping denosumab causes rapid rebound bone loss/vertebral fractures, so it cannot just be abruptly discontinued.

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12. Phenytoin/phenobarb ↓Ca

WHAT YOU SEE

Vampire figures, induce CYP, inactivate vit D

WHAT IT ENCODES

Phenytoin and phenobarbital are potent CYP450 INDUCERS that accelerate hepatic catabolism of vitamin D to inactive metabolites → vitamin D deficiency, hypocalcemia, and osteomalacia/rickets with chronic use. Patients on long-term enzyme-inducing anticonvulsants need vitamin D monitoring and supplementation.

BOARD TRAP

WRONG: missing osteomalacia in a chronic epileptic. Bone pain, low calcium, low 25-D, and high alk phos in a patient on phenytoin/phenobarbital is drug-induced vitamin D deficiency — supplement vitamin D, don't just chase the calcium.

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13. PPIs ↓Ca

WHAT YOU SEE

PPI pills, chronic use low absorption

WHAT IT ENCODES

Chronic proton-pump inhibitors raise gastric pH and reduce solubility/absorption of calcium carbonate (and magnesium) → hypocalcemia and increased fracture risk over years. PPI-induced hypomagnesemia can ALSO worsen calcium by impairing PTH secretion/action.

BOARD TRAP

WRONG: repleting calcium alone in a long-term PPI user with hypocalcemia. Check and correct MAGNESIUM too — PPI hypomagnesemia causes refractory hypocalcemia that won't respond until Mg is restored.

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14. Citrate ↓Ca

WHAT YOU SEE

Ca citrate jug, chelation, transfusion

WHAT IT ENCODES

Citrate (anticoagulant in MASSIVE TRANSFUSION, apheresis, and regional citrate anticoagulation for CRRT) CHELATES ionized calcium → acute hypocalcemia with perioral tingling, tetany, and prolonged QT, while TOTAL calcium may look normal. Liver dysfunction worsens it (citrate not metabolized). Treat by repleting calcium and monitoring ionized Ca.

BOARD TRAP

WRONG: trusting total calcium during massive transfusion. Citrate drops the IONIZED fraction specifically — total Ca can be normal/high while the patient is symptomatically hypocalcemic. Always monitor IONIZED calcium in these settings.

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15. Stop offending agent first

WHAT YOU SEE

Bartender sign: STOP THE OFFENDING AGENT FIRST

WHAT IT ENCODES

Core management principle for ALL drug-induced calcium disorders: identify and STOP the culprit before escalating therapy. Often discontinuation (± hydration) is curative for thiazide/lithium hypercalcemia, and repleting the missing factor (vitamin D, magnesium, calcium) reverses drug-induced hypocalcemia. Reconcile the full med list at the bedside.

BOARD TRAP

WRONG: stacking calcium-lowering drugs (saline, calcitonin, bisphosphonate) onto a patient whose hypercalcemia is simply from a thiazide. You may overshoot into hypocalcemia — withdraw the offending agent and reassess first.
